Step 1 Producer Authentication & Registration

Producer registration includes

→ unique producer identifier ($PID_p$)

→ Public - private key pair using ECC

→ storage of producer's public key within the block-chain ledger

$$PID_p = H(PK_p \| T_p)$$

$T_p$ → timestamp

$PK_p$ → publick key

$H$ → Hash function SHA - 256

Step 2 Secure chunk Generation and Cryptographic authentication

(i) Content $C = \{C_1, C_2 \dots C_k\}$

(ii) a symetric AES - 256 session key generate. (SK)

(iii) for each chunk is encrypted using

$$E_i = AES_{SK}(C_i)$$

(iv) Integrity of encrypted chunk is using SHA - 256

$$H_i = SHA(E_i)$$

(v) producer generate digital signature using ECC

$$Sig_i = ECC_{sig}(H_i, SK_{priv})$$

$SK_{priv}$ → producer private signing key step 3 Blockchain - Based Metadata Registration

Metadata Recorded on the ledger for minimizing the storage requirement of blockchain

Blockchain registration record for chunk i

$$BR_i = \{ \underbrace{CID_i}_{\text{Chunk Identifier}}, \underbrace{H_i}_{\text{chunk Hash}}, \underbrace{PID_p}_{\text{producer identifier}}, \underbrace{T_i}_{\text{Timestamp}} \}$$

step 4 Content received through multiple transmission paths.

* Consumer verifies each chunk independently

$$H_i' = SHA256(F_i)$$

Corresponding blockchain Record

$$H_i^{BC} = lookup(CID_i)$$

Content authenticity is verified as

$$V_i = \begin{cases} 1 & H_i' = H_i^{BC} \\ 0 & \text{otherwise.} \end{cases}$$

If verification succeeds, the producer signature is valided using public key

$$ECC_{verify}(Sig_i, H_i, PK_p) = True$$

Prevents :→ ① cache poisoning attacks
② content tampering
③ Malicious producer impersonation
④ Replay Attack.

Step 5. Security Overhead optimization

① Hybrid Cryptographic Architecture

#. AES-256 ⇒ for data encryption

# ECC → authentication

# SHA-256 → integrity varification

~~Cryptografic Cost~~

~~Cryptographic Cost~~